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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I Semester B. E. Examinations May-2023

Common to AI / AS/ BT / CH / CS / CY / CD / EC/EI/ET/ IM / IS / ME / CV

BASICS OF ELECTRICAL ENGINEERING (ELECTIVE)

Time: 03 Hours *Maximum Marks: 100*

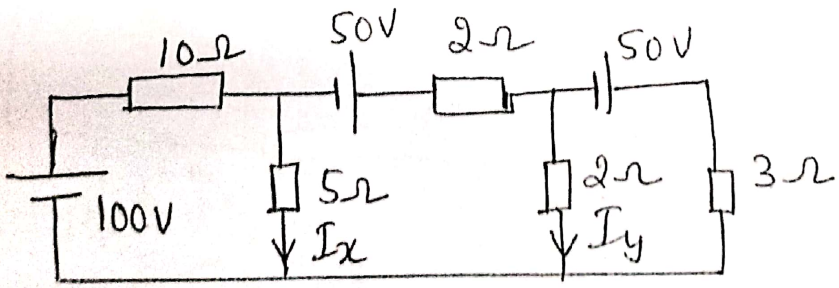
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. Question number 2 is compulsory. Choose any one full question from 3 or 4, 5 or 6, 7 or 8 and 9 or 10.

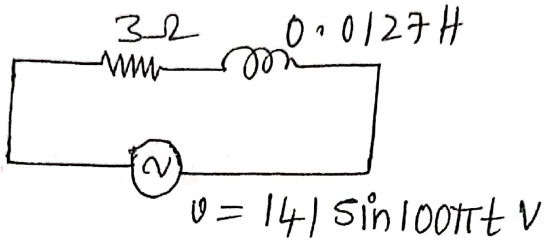
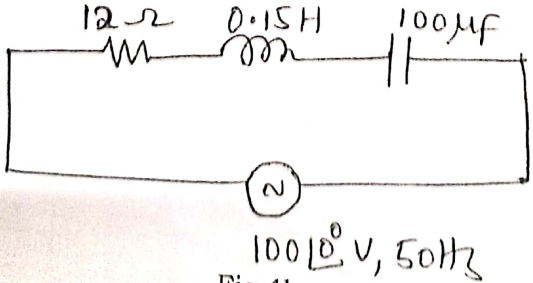
PART-A

1	1.1	Define Kirchhoff's laws with illustrative example.	02
	1.2	A heater is given a current of $10A$ from a $250V$ source for $15hours$. Determine the energy consumed by the heater in kWh.	02
	1.3	Define RMS and average value.	02
	1.4	A series circuit with $R = 10\Omega$, $L = 50mH$ and $C = 100\mu F$ is supplied with $200V, 50Hz$. Find impedance value in the circuit.	02
	1.5	Write the relation between phase and line values of voltage and currents in star and delta connected loads.	02
	1.6	Define efficiency and regulation of a transformer.	02
	1.7	A $3-$phase, $4-$pole, $400V, 50H$. Induction motor runs with a speed of $1440rpm$. Calculate its slip.	02
	1.8	List the types of single-phase induction motor.	02
	1.9	Define fuse and miniature circuit breaker.	02
	1.10	Write any four safety precaution to avoid electric shock.	02

PART-B

2	a	State and prove Thevenin's theorem.	08
	b	Using KCL and KVL, determine the currents I_x and I_y in the network shown in Fig 2b.  Fig 2b	08
3	a	Show that the power consumed in an RC series circuit is $VI\cos\phi$. Draw the waveform for voltage, current and power.	06

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	b	Define: i) Instantaneous value ii) Amplitude iii) Cycle iv) Period With respect to sinusoidally varying quantities.	
	c	For the series RL circuit shown in Fig 3c, i) Calculate rms value of current and its angle ii) Expression for current iii) Average power dissipated in the circuit iv) Determine pf v) Draw the phasor diagram.  Fig 3c	06
		OR	
4	a	An alternating current of frequency $60Hz$ has a maximum value of $12A$. i) Write down the equation for its instantaneous value. ii) Find the value of current after $\frac{1}{360} seconds$ iii) Find the time taken to reach $9.6A$ for the first time.	06
	b	For the circuit shown in Fig 4b, calculate i) The impedance ii) The current iii) Phase angle iv) Voltage across each element v) Power factor vi) Apparent power vii) The average power. Also, draw the phasor diagram for the circuit.  Fig 4b	06
5	a	Obtain the relationship between line current and phase current in a balanced 3ϕ delta connected system.	10
	b	What are the advantages of 3ϕ systems over a single phase system?	06 04

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	c	Two-wattmeter method was used to determine the input power to 3ϕ motor. The reading were $5.2kW$ and $-1.7kW$ and the line voltage was $415V$. Calculate: i) Total power ii) Power factor iii) Line current.	06
		OR	
6	a	Explain the principle of operation of a single phase transformer and derive its EMF equation.	08
	b	A $600kVA$ transformer has an efficiency of 92% both at full load unity pf and half load $0.9pf$. Determine its efficiency at 75% of full load and $0.9pf$.	08
7	a	Describe the constructional differences between a squirrel cage rotor and wound rotor of an induction motor. Discuss their relative advantages and disadvantages.	06
	b	Define slip. Derive an expression for the slip and frequency of the rotor current.	05
	c	A 6 –pole induction motor is supplied by a 10 –pole alternator which is driven at $600rpm$. If the motor is running at $970rpm$, determine the percentage slip.	05
		OR	
8	a	Explain the concept of rotating magnetic field of an induction motor.	10
	b	Explain and draw torque-slip characteristics of a 3ϕ induction motor. Describe the construction of a single phase induction motor with the aid of diagram.	06
9	a	With the help of a block diagram, explain the concept of power transmission and power distribution.	08
	b	Explain with an example, how the electricity bill be generated for domestic consumers.	08
		OR	
10	a	Explain the working principle of fuse and MCB and also explain the merits and demerits of each one.	06
	b	What is earthing and explain with diagram pipe earthing.	06
	c	What are the safety precautions to be taken to avoid electric shock?	04

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I / II Semester B. E. Examinations Oct/Nov 2023

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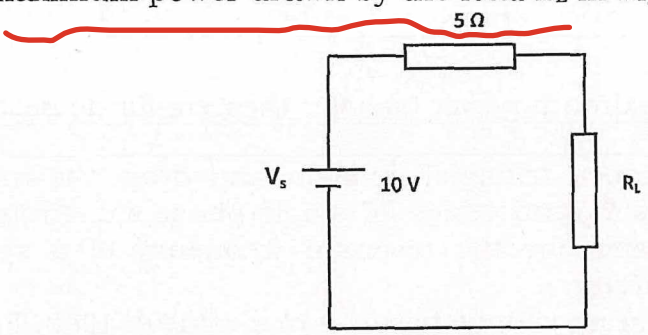
BASICS OF ELECTRICAL ENGINEERING

Time: 03 Hours *Maximum Marks: 100*

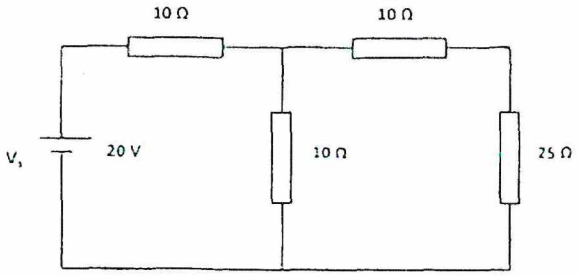
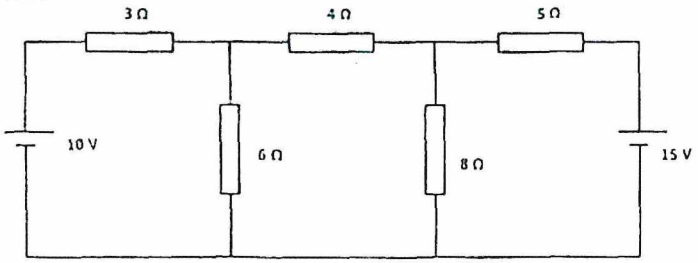
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. Question number 2 is compulsory. Choose any one full question from 3 or 4, 5 or 6, 7 or 8 and 9 or 10.

PART-A

1	1.1	The maximum power drawn by the load R_L in Fig. 1.1 will be?  Fig. 1.1	02
	1.2	The resistance of the two coils is 25ohms when connected in series, and 6 ohms when connected in parallel. Determine the individual resistances of the two coils.	02
	1.3	Define a) Form factor b) Peak factor.	02
	1.4	A $50-Hz$ sinusoidal current has peak factor 1.4 and form factor 1.1. Its average value is $20A$. The instantaneous value of current is $15A$ at $t = 0$. Write the equation of current.	02
	1.5	In a balanced 3-phase system, power is measured by two wattmeters and the ratio of two wattmeter readings is 2:1. Determine the power factor of the system.	02
	1.6	State the differences between a core-type and a shell-type transformer.	02
	1.7	Why the rotor slots in a three-phase induction motor are purposely given a slight skew?	02
	1.8	List the types of single-phase induction motor.	02
	1.9	Mention the characteristics of fuse.	02
	1.10	What is an electric shock? Write safety precautions to avoid electric shock.	02

PART-B

2	a	Use Thevenin's theorem to determine the current through and the voltage across the 25Ω resistor given in Fig. 2a.  Fig. 2a	06
	b	A network is arranged as shown in Fig. 2b. Determine the current in each resistance.  Fig. 2b	06
	c	State and prove maximum power transfer theorem for dc networks.	04
3	a	What is an impedance triangle? Explain and draw the impedance triangle for a series *RL* and series *RC* single-phase a.c. circuits. Also, deduce an expression for the resonant frequency of a series *RLC* single-phase a.c. circuit.	08
	b	A sinusoidal alternating voltage has an r.m.s value of $100V$. Find the i) Instantaneous value of $0.0125s$ after passing through maximum positive value and ii) The time measured from a positive maximum value when the instantaneous voltage is $70.7V$.	08
		OR	
4	a	Derive the expression for Average value and rms value of a sinusoidal waveform.	08
	b	A coil and a non-inductive resistor are connected in series across a $200V, 50\ Hz$ supply. The voltage across the coil and resistor are $120V$ and $140V$ respectively. If the supply current is $0.5A$, calculate: i) The resistance and inductance of the coil ii) The power dissipated in the coil iii) The power factor of the coil iv) The power factor of the circuit.	08
5	a	The three arms of a three-phase load each comprise an inductor of resistance 25Ω and of inductance $0.15\ H$ in series with a $120\mu F$ capacitor. The supply is $415\ V, 50\ Hz$. Calculate the line current and the total power in watts, when the three arms are connected in: i) star ii) delta.	08

	b	Give reasons: i) Copper loss is called as variable loss and iron loss is called as Constant loss. ii) The rating of transformer is in kVA. iii) Transformer cannot be excited by DC supply.	08
		OR	
6	a	A single-phase transformer working at unity power factor has an efficiency of 90% at both half load and the full-load of $500W$. Determine the efficiency at 75% full load and the maximum efficiency.	08
	b	Derive the relationship between the line and phase values of voltage in a three-phase start connected system with the aid of a phasor diagram.	08
7	a	Explain the concept of rotating magnetic field of an Induction motor. Draw the torque-slip characteristics of a three-phase induction motor.	08
	b	Describe the construction of a single-phase induction motor with the aid of a diagram.	08
		OR	
8	a	Explain the working principle of a single-phase induction motor. Discuss why single-phase induction motors do not have a starting torque.	08
	b	Compare between Squirrel cage and slip ring induction motor. A 12-pole, 3-phase alternator is coupled to an engine running at 500 r.p.m. It supplies an induction motor which has a full-load speed of 1440 r.p.m Find the slip and the number of poles of the motor.	08
9	a	In a residential house, the following load are connected: i) Six lamps of $40W$ each, switched on for 5 hours a day. ii) Two fans of $60W$ each, switched no for 12 hours a day. iii) One $100\ W$ heater working for 2 hours per day. iv) One refrigerator of $250W$ working for 10 hours per day. If each unit of energy costs $Rs.1.90$, what will be the total cost in the month of September?	08
	b	Illustrate the concept of power generation, transmission, and distribution system with block diagrams.	08
		OR	
10	a	What are the types of earthing? Explain with diagram any one type of earthing.	08
	b	Explain the working of Miniature Circuit Breaker with neat diagram. Mention its merits and demerits.	08

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I / II Semester B. E. Regular / Supplementary Examinations Feb-2024

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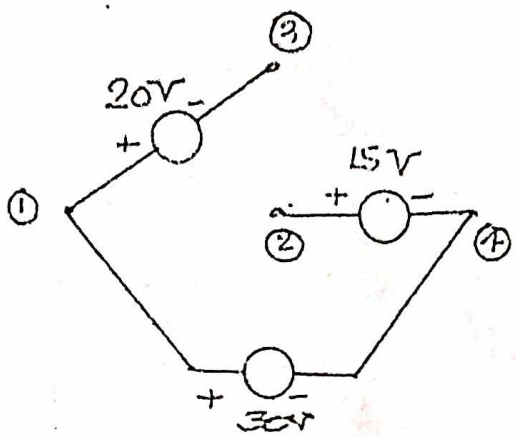
BASICS OF ELECTRICAL ENGINEERING

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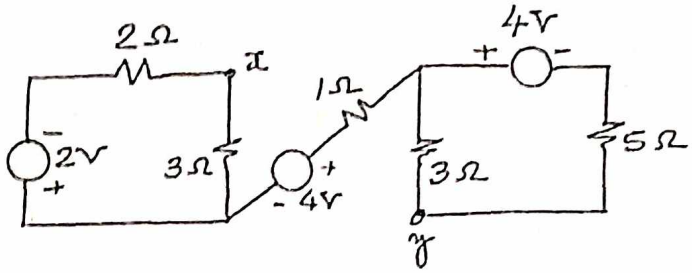
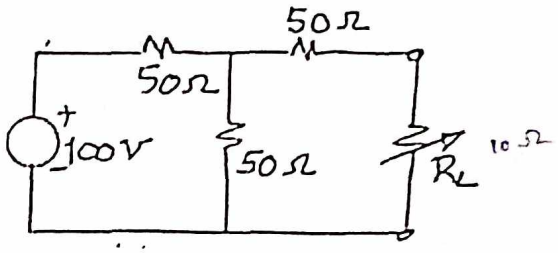
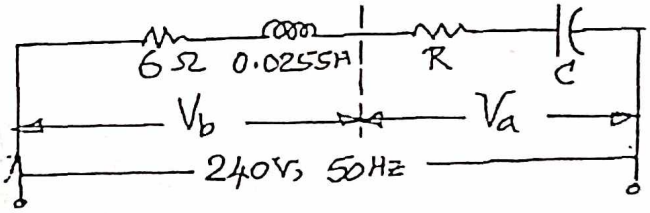
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer SIX full questions from Part B. In Part B question numbers 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.

PART-A

1	1.1	Refer to Fig. 1.1 and find the voltages V_{12}, V_{23} and V_{34}.  Fig. 1.1	02
	1.2	An AC voltage of $e = 200\sin(314t - 30^0)$ $Volts$ drives a current of $i = 20\sin(314t + 30^0)$ $Amperes$. What will be the power factor and the average power?	02
	1.3	The necessary two conditions of a balanced three phase vectors are _____.	02
	1.4	What is hysteresis loss and how it is minimized in transformers?	02
	1.5	A $3-phase, 4-pole, 400V, 50Hz$ induction motor runs at a speed of $1440rpm$, calculate its slip.	02
	1.6	The vectorial sum of three balanced currents is _____.	01
	1.7	Mention the different types of single phase transformers.	01
	1.8	Why the core loss is considered as constant loss in transformers?	01
	1.9	What is the condition for maximum efficiency in a transformer?	01
	1.10	What do you mean by slip?	01
	1.11	The average value of a pure cosine wave having amplitude of $10V$ is _____.	01
	1.12	The rms value of a sine wave is $10units$, what is peak value?	01
	1.13	Define voltage regulation of a transformer.	01
	1.14	Why the efficiency of a transformer is always higher than other machines?	01
	1.15	Why the induction motors are also called asynchronous motors?	01

PART-B

2	a	In the circuit shown in Fig. 2.a, find the potential difference across $x-y$ terminals and the power delivered by the 1 *Ohm* resistor.  Fig. 2.a	08
	b	State and explain Thevenin's theorem, using this theorem find the power dissipated to a load resistance of 10 *Ohms* in the circuit shown in Fig. 2.b  Fig. 2.b	08
3	a	Show that the average power consumed in an *AC* circuit in $VI\cos\phi$.	06
	b	Show that the average power consumed by a pure inductor is zero.	04
	c	An alternating current is given by : $i = 10\sin 942t\ A$. Determine: i) frequency, ii) the time taken from $t = 0$ for the current to reach a value of $6A$ for the first and second time, iii) the energy dissipated when the current flows through a 20 *Ohm* resistor for 30 *minutes*.	06
		OR	
4	a	Draw and explain the vector diagram of an $R-L-C$ series circuit excited by a supply of E volts when: i) $X_L = X_C$, ii) $X_L > X_C$, iii) $X_L < X_C$.	08
	b	Find the value of R and C in the circuit shown in Fig. 4.b, so that $V_b = 3V_a$ and V_b are in quadrature. Also find the current I.  Fig. 4.b	08

5	a	Write the mathematical representation of three phase voltages with i) instantaneous values, ii) *RMS* values, iii) vector diagrams.	08
	b	Show that two watt meters are sufficient to measure three phase power with a delta connected load.	08
		OR	
6	a	Show that the $e.m.f.$ induced per turn is same for both primary and secondary windings of transformers.	08
	b	The primary and secondary windings of a $500kVA$ transformer have resistances of 0.42 and 0.0019 Ohm respectively. The primary and secondary voltages are 11000 and 400V respectively and the core loss is $2.9kW$. Assuming the power factor of 0.8 calculate the efficiency at full load.	08
7	a	Explain the principle of torque production in three phase induction motors.	05
	b	Draw the typical torque slip characteristics of wound rotor induction motor and mark all the salient points on it.	05
	c	A $6-pole$ induction motor is supplied by a $3-phase, 50Hz$ supply has a rotor frequency of 2.3 Hz. Calculate: i) the percentage slip, ii) speed of the rotor. What will be these values if the stator is wound for 4 poles?	06
		OR	
8	a	Explain why the starting torque is zero for a single phase induction motor, and how this will be produced.	05
	b	Draw the electrical schematics of various types of single phase induction motors.	05
	c	Draw and explain the rotor construction of the two types of three phase induction motors.	06
9	a	Define the term 'Power System' and explain the same with the help of a block diagram showing all its components.	06
	b	Differentiate between 'Fuse' and '*MCB*' and mention the advantages and disadvantages.	06
	c	What are preventive measures of electrical shock, explain.	04
		OR	
10	a	What is earthing, why it is necessary and explain with diagram 'plate earthing'?	08
	b	A domestic house uses the following appliances whose details are in the table below:	08

Sl. No.	*Appliances*	*Power Rating*	*Quantity Nos.*	*Usage per day (Hrs)*
1	*LED bulbs*	9 *W*	10	4
2	*LED tube lights*	20 *W*	3	6
3	*Geysers*	2 *kW*	2	1
4	*Water pump*	500 *W*	1	2
5	*Ceiling fans*	55 *W*	6	3
6	*Mixer grinder*	1200 *W*	1	1
7	*Induction top*	1500 *W*	2	6

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I / II Semester B. E. Regular / Supplementary Examinations Aug-2024

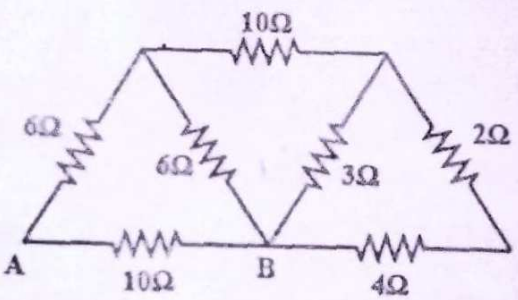
BASICS OF ELECTRICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

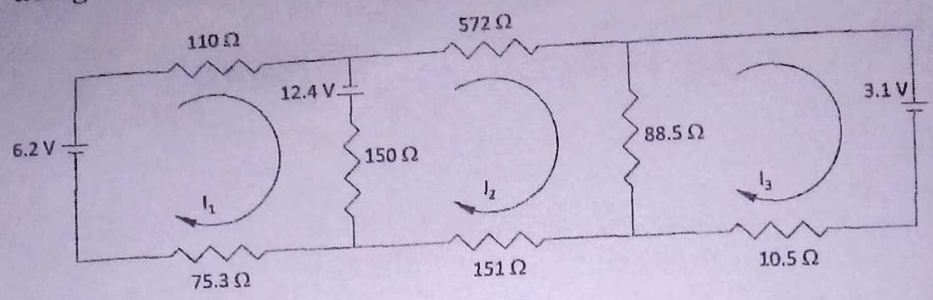
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1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define Ohm's Law?	01	1	1
	1.2	What is the resistance of a good insulator?	01	3	2
	1.3	Determine the equivalent resistance between AB of the following circuit shown in Fig. 1.3.  Fig. 1.3	02	3	2
	1.4	List out any two advantages of AC system over DC system.	01	2	1
	1.5	Define Form Factor.	01	1	1
	1.6	The complex Volt Amperes in a series circuit are $(4330 - j2500)$, and the current is $(25 + j43.3)$ A. Find the applied voltage.	02	3	2
	1.7	The power flowing in a 3 –phase, 3 –wire balanced load system is measured by two wattmeter method. The reading in a wattmeter A is 750 W and wattmeter B is 1500 W. What is the system's power factor?	02	3	2
	1.8	What are the different types of losses in transformer and which tests are conducted to calculate the losses?	02	1	1
	1.9	A 3-Phase, 4 pole, 440 V, 50 Hz induction motor runs with a slip of 5%. Find the rotor speed and frequency of the rotor current.	02	3	2
	1.10	Draw the Torque-slip curve for three-phase induction motor.	02	3	1
	1.11	Write down the advantages and disadvantages of Fuse.	02	2	1
	1.12	A geyser is rated at 2 kW, 230 V, 50 Hz. If it is switched ON for one hour daily, what would be the energy cost saving, at the rate of $Rs.$ 2.50 per unit if it is replaced by a solar water heater?	02	3	4

PART-B

2	a	State Maximum Power Transfer Theorem and prove that maximum power transfer when $R_L = R_{TH}$.	08	4	1
	b	Find the currents I_1, I_2 and I_3 in the circuit shown in Fig. 2b using *KVL/KCL*.  Fig. 2b	08	3	2
3	a	With neat sketch, phasor diagram and necessary waveforms, prove that average power in a pure inductive circuit is zero.	06	1	1
	b	A series *RLC* circuit containing a resistance of 12 Ω, an inductance of 0.15 *H* and a capacitor of 100 μ*F* are connected in series across a 100 *V*, 50 *Hz* supply. Calculate the total circuit impedance, voltage across each element, the circuit current, power factor, actual power consumed, reactive power and draw the voltage phasor diagram.	07	3	2
	c	i) What is resonance condition? ii) What is the power factor at resonance? iii) What is the reactive power at resonance?	03	1	2
		OR			
4	a	With a neat circuit, phasor diagram and waveforms deduce the equations for voltage, current, phase angle and average power for a series *RLC* circuit in different cases: i) $X_L > X_C$; ii) $X_L < X_C$; iii) $X_L = X_C$.	08	1	2
	b	Define: i) Average value ii) *RMS* value iii) Form Factor iv) Peak factor	04	1	1
	c	Derive Average value, *RMS* value, Form factor and Peak factor for sinusoidal current.	04	3	1
5	a	In a three-phase balanced circuit using two-wattmeter method, derive power factor, active power, reactive power and apparent power for the following connections: i) Balanced star-connected load ii) Balanced delta-connected load	10	1	2
	b	A balanced three-phase delta-connected load has per-phase impedance of $(25 + j40)\Omega$. If 400 *V*, 3-phase supply is connected to this load, find: i) phase current ii) line current iii) power supplied to the load.	06	3	1
		OR			

6	a	Prove that the efficiency of the transformer will be maximum when copper losses equal to constant losses.	06	3	1
	b	A 600 kVA transformer has an efficiency of 92% at full load, unity p.f. and half full load, 0.9 $p.f$. Determine its efficiency at 75% of full load and 0.9 $p.f$.	08	3	2
	c	Compare core type and shell type transformers.	02	1	2
7	a	With a neat sketch and vector diagram, mathematically explain how the rotating magnetic field is produced in the three-phase induction motor.	08	2	2
	b	A 4 pole, 50 Hz induction motor has a slip of 1% at NO load. When operated at full load, the slip is 2.5%. Find the change in speed from no load to full load.	04	3	3
	c	Distinguish between Squirrel cage and Slip ring induction motor.	04	2	2
		OR			
8	a	Explain the concept of double-revolving field theory.	08	1	2
	b	Name the different types of single-phase induction motors and applications.	05	1	2
	c	A 10-pole induction motor is supplied by a 6-pole alternator, which is driven at 1200 rpm. What is the speed if the induction motor runs with a slip of 3%?	03	3	3
9	a	With the help of a block diagram, explain the concept of power transmission and power distribution.	08	1	3
	b	Estimate the total daily energy requirement for the loads shown in the table below. Also, calculate monthly electricity bill if the electricity cost is $Rs.2$/unit for the first 40 units, $Rs.4$/unit for next 50 units and $Rs.6$/unit for rest of the units.	08	5	4
		OR			
10	a	Define Earthing, what is the necessity of earthing and types. Explain in detail the plate earthing with neat sketch.	08	1	4
	b	What are the safety precautions to be taken to avoid electric shock?	04	5	4
	c	Differentiate between Fuse and MCB.	04	3	1

Name of the appliance	*Power Rating (W)*	*Average daily usage hrs.*	*Number of appliances*
Fan	50	10	4
CFL	12	6	8
LED TV	100	5	2
Refrigerator	300	24	1
Laptop	60	5	1

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I/II Semester B. E. Supplementary Examinations May-2024

Common to all Branches

ELEMENTS OF ELECTRICAL ENGINEERING

Time: 03 Hours *Maximum Marks: 100*

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2. Answer FIVE full questions from Part B. Question number 2 is compulsory. Choose any one full question from 3 or 4, 5 or 6, 7 or 8 and 9 or 10.

PART-A

			M	BT	CO
1	1.1	Define renewable and non-renewable energy sources.	02	1	3
	1.2	What is synchronous generator? What is the principle of operation of synchronous generator?	02	1	1
	1.3	List the raw material required for power generation in case of hydel, thermal, nuclear and *PV* power plants.	02	1	1
	1.4	What do you mean by lagging power factor and leading power factor?	02	1	2
	1.5	List any four advantages of three phase system.	02	1	2
	1.6	What do you mean by 1*unit* of electricity?	02	1	2
	1.7	What is the difference between fuse and *MCB*?	02	1	1
	1.8	What is the difference between transformer and auto transformer?	02	1	3
	1.9	Define slip of induction motor.	02	1	1
	1.10	Why DC series motor should never be started without any load on it?	02	2	3

PART-B

2	a	With neat sketch, explain the operation of Hydel power plant.	06	1	1
	b	Explain $I-V$ and $P-V$ characteristics of a PV cell.	06	1	4
	c	What is a Battery? Why batteries are connected in: i) Series, ii) Parallel	04	2	2
3	a	With circuit and phasor representation obtain current, voltage and power relations in a series *RL* crcuit.	06	2	2
	b	Define the following terms: i) Real power ii) Reactive power iii) Apparent power, and iv) Power factor	04	1	2
	c	A delta connected load consists of a resistance of 10Ω and a capacitance of $100\mu F$ in each phase. A supply of $410V$ at $50Hz$ is applied to the loads. Find the line current, power factor and power consumed by the loads.	06	1	2
		OR			

4	a	Derive the relation between phase and line voltages and current in a star connected system.	06	4	2
	b	With phasor representation show that Two wattmeters are sufficient to measure the power in a three phase system.	06	2	2
	c	Find an expression for the current and calculate the power, when a voltage of $e = 283\sin(100\pi t)$ is applied to a coil having $R = 50\Omega$ and $L - 0.159\,H$.	04	1	2
5	a	What are the different types of Electric wiring schemes? Explain with diagram any two types.	06	1	1
	b	Discuss the power ratings of different household appliances.	04	2	1
	c	List the merits and demerits of fuse, *MCB* and *ELCB*.	06	1	1
		OR			
6	a	What is earthing? Why earthing is required? Explain anyone type of earthing.	06	1	1
	b	With block diagram, explain the working of Uninterrupted power supply.	04	1	2
	c	Estimate Total Daily Energy Requirement for the following loads:			

Name of the appliance	*Power Rating (W)*	*Avg. Daily Usage Hrs*	*No. of Appliances*
CFL	12	6	4
Fan	50	8	2
TV (21")	150	2	1
Computer	250	3	1
Water heater	1000	2	1

		Take electricity cost to be *Rs.* 5 per unit.	06	2	2
7	a	What is a transformer? From basics, derive *EMF* equation of transformer.	06	1	3
	b	Discuss about the losses that occur in a transformer. Hence write efficiency equation of the transformer.	04	2	3
	c	A $250kVA$, single phase transformer has an efficiency of 96% on full load $0.8pF$ lagging and at half $0.85pF$ lagging. Find the iron loss and full load copper loss.	06	1	3
		OR			
8	a	Explain how rotating magnetic field is generated in a $3-phase$ induction motor.	06	1	3
	b	Discuss about the different types of rotors used in a three phase induction motor.	04	2	3
	c	A 4 pole $50Hz$, induction motor has a slip of 1% at no load. When operated at full load, the slip is 2.5%. Find the change in speed from no load to full load.	06	2	3
9	a	With a cross sectional view, explain the construction and operation of a *DC* motor.	06	1	3
	b	Draw and explain torque v/s current and speed v/s current characteristics of *DC* shunt motor.	04	1	3

	c	A 4 *pole* 220*V* lap connected *DC* shunt motor has 36 *slots*, each slot containing 16 conductors. It draws a current 40 *amps* from the supply. The field resistance and armature resistance are 110Ω and 0.1Ω respectively. The motor develops an output power of 6*kW*. The flux per pole is 40*mWb*. Calculate the: i) Speed ii) The torque developed by the armature, and iii) The shaft torque.	06	2	3
		OR			
10	a	What is the significance of back *EMF* in a *DC* motor?	04	1	3
	b	Derive an equation for the torque developed in the armature of a *DC* motor.	06	4	3
	c	Explain the construction and working principle of stepper motor.	06	1	3

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Supplementary Examinations Oct-2023

ELEMENTS OF ELECTRICAL ENGINEERING

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Choose any one full question from 3 or 4, 5 or 6, 7 or 8 and 9 or 10.

PART-A

1	1.1	Name two non-conventional source of energy that is used most in India and why?	02
	1.2	A $3-phase, 50Hz, 6$ pole star connected alternator has 360 conductors per phase and flux per pole is $0.5Wb$. Calculate the e.m.f generated if the winding factor is 0.97 and the coils are full pitched.	02
	1.3	What are the components of a solar PV standalone system?	02
	1.4	What is power factor? Why power factor improvement is required?	02
	1.5	A balanced 3-phase start connected load draws power from 400V supply. The two wattmeter's read $3kW$ and $2kW$. Calculate line current in the circuit.	02
	1.6	What is the difference between MCB and $ELCB$?	02
	1.7	A $50kVA, 1000/200V$ transformer takes $4A$ at $0.2\,p.f$ when the secondary is open. Determine core loss and magnetizing component of current.	02
	1.8	What is slip of an induction motor? What is the value of the slip when the motor is at standstill?	02
	1.9	Why a shunt motor should not be turned on with its field winding open?	02
	1.10	What are the advantages of $BLDC$ motor over induction motors?	02

PART-B

2	a	With a neat sketch, explain the constructional features of smooth cylindrical rotor alternator?	05
	b	Explain the following. i) I-V and P-V characteristics of solar PV ii) Block diagram of smart grid.	08
	c	A $12V$ battery of capacity $500Ah$ is connected to a solar panel. If battery discharge duration is $10hours$, then calculate the power of the battery and energy stored in the battery.	03
3	a	Derive an expression for the current drawn by a pure inductor. Explain with the help of a power diagram that the value of an average power drawn by the inductor during one cycle is zero.	05

	b	An alternating voltage is represented by the equation $v = 100\sin 520t\ V$. Calculate its: i) rms value ii) average value iii) form factor iv) peak factor v) frequency vi) voltage when $t = 0.0015\ sec$.	06
	c	With phasor representation and circuit diagram, derive an expression for measurement of power by two-wattmeter method.	05
		OR	
4	a	Derive the power dissipated in a series RC circuit and also draw the waveforms of voltage, current and power.	05
	b	Three identical coils are connected in a star to a $400V, 3-phase, 50Hz$ supply and each coil takes $300W$. If the power factor is 0.8, calculate i) line current ii) impedance iii) resistance and inductance of each coil.	06
	c	Prove that in a balanced star connected system, the line voltage is 1.732 times of phase voltage and the line current is same as phase current.	05
5	a	A household uses the following electric appliances: i) refrigerator of rating $400W$ for 24hours each day ii) two electric fans of rating $80W$ each for 12hours each day iii) six electric bulbs of $18W$ for 6hours each day Calculate the electricity bill of the household for the month of April if the cost per unit of electric energy is $Rs.3.50$.	05
	b	Explain the merits and demerits if conduit wiring.	05
	c	With a neat sketch, explain the process of plate earthing.	06
		OR	
6	a	With the help of block diagram, explain the working of UPS.	06
	b	Define the term 'Fuse' and mention the requirements for a good fuse. Also explain how fuse is different from MCB.	05
	c	Estimate Total Daily Energy Requirement for the following loads and also compute the electricity bill for the month of July assuming average cost $Rs.4.50$ per unit.	05

Name of the appliances	*Power rating (W)*	*Average daily usage (Hrs.)*	*No. of appliances*
CFL	25	7	4
FAN	50	6	2
TV	150	6	1
COMPUTER	200	5	1
GYSER	2000	1	1

7	a	Explain the working of capacitor start capacitor run 1-phase induction motor. Also, draw the phasor diagram.	06
	b	A $50kVA$ transformer has an efficiency of 98% at full load, $0.8p.f$ and an efficiency of 96.9% at one fourth of full load, $0.8p.f$. Determine i) The iron loss and full load copper loss ii) Losses at maximum efficiency iii) Fraction of full load corresponding to maximum efficiency.	06
	c	The slots on the rotor of an induction motor are usually skewed. Justify.	04
		OR	
8	a	With a neat sketch, explain the constructional features of 1-phase shell type transformer.	06
	b	A $3-phase$, $50Hz$ induction motor runs at $965\ RPM$ at full load. Calculate i) The number of motor poles ii) The slip and frequency of the rotor currents iii) The speed of the stator field with respect to rotor and with respect to the rotor field.	06
	c	Core type transformer is used for high voltage application. Justify.	04
9	a	Explain the constructional features, working principle and applications of DC motor.	09
	b	With a neat circuit, explain construction and working principle of stepper motor.	07
		OR	
10	a	Explain the characteristics T_a/I_a and N/I_a of DC series and shunt motor. Justify the nature of the curves.	09
	b	With a neat circuit, explain the construction and working principle of $BLDC$ motor.	07